

Learning Day 6 - Supplementary Notebook

Autoencoder-Based Anomaly Detection for Financial Transactions

Master Documentation

Project focus

Use an autoencoder to learn normal transaction behaviour and flag suspicious transactions using reconstruction error. The fraud label is used only for evaluation, not as a training target.

Item	Details
Notebook	anomaly_detection_autoencoder.ipynb
Dataset	PaySim Synthetic Financial Datasets for Fraud Detection, Kaggle handle: ealaxi/paysim1
Business domain	Financial transaction fraud / suspicious transaction detection
Model type	Dense autoencoder trained on normal transactions
Decision signal	Reconstruction error
Selected threshold	95th percentile of normal validation reconstruction error

Table of Contents

1. 1. Executive Summary
2. 2. Business Problem and Dataset
3. 3. Theory A-Z
4. 4. Notebook Workflow Flowchart
5. 5. Variable and Method Flow
6. 6. Data Preparation
7. 7. Model Architecture and Training
8. 8. Results and Chart Interpretation
9. 9. Thresholding and Evaluation
10. 10. Business KPI Summary
11. 11. Limitations and Next Steps
12. 12. Interview-Style Explanation
13. 13. Glossary

1. Executive Summary

This master document explains the anomaly detection notebook built for financial transaction fraud analysis using an autoencoder. The model was trained only on normal transactions and then used reconstruction error as an anomaly score on a test set containing both normal and fraud transactions.

Main result

Fraud transactions had much higher reconstruction error than normal transactions. The selected 95th percentile threshold produced fraud precision of 0.868 and fraud recall of 0.605.

Metric	Value	Meaning
Full dataset fraud rate	0.1291%	Fraud is extremely rare in the original data.
Working dataset	108,213 rows	100,000 normal transactions + all 8,213 fraud records.
Input dimension	11	6 scaled numeric features + 5 encoded transaction-type columns.
Final training loss	0.013953	Reconstruction loss on normal training data.
Final validation loss	0.013549	Reconstruction loss on normal validation data.
Selected threshold	0.098729	95th percentile of validation reconstruction error.
Detected fraud (TP)	4,968	Fraud transactions correctly flagged as suspicious.
Missed fraud (FN)	3,245	Fraud transactions not detected at selected threshold.

2. Business Problem and Dataset

2.1 Business Problem

Financial fraud detection is a rare-event problem. Most transactions are normal, while only a small fraction are fraudulent or suspicious. This makes simple accuracy misleading because a model can appear strong by predicting almost everything as normal.

- Fraud cases are rare compared with normal transactions.
- Manual review is expensive and cannot cover every transaction.
- Suspicious behaviour can change over time.
- A useful model should create a risk signal that supports fraud investigation.

2.2 Dataset Summary

Column	Role in Notebook	Explanation
step	Feature	Transaction time step.
type	Feature	Transaction type such as PAYMENT, TRANSFER, CASH_OUT, CASH_IN, DEBIT.
amount	Feature	Transaction amount.
oldbalanceOrig	Feature	Sender balance before transaction.
newbalanceOrig	Feature	Sender balance after transaction.
oldbalanceDest	Feature	Receiver balance before transaction.

newbalanceDest	Feature	Receiver balance after transaction.
isFraud	Evaluation label	Used only after predictions to evaluate performance.
isFlaggedFraud	Excluded	Existing rule-based fraud flag, excluded to avoid leakage.
nameOrig / nameDest	Excluded	Sender and receiver IDs, not used directly as model inputs.

2.3 Fraud Imbalance

Class	Transactions	Share
Normal Transaction	6,354,407	99.8709%
Fraud Transaction	8,213	0.1291%

Fraud represents only 0.1291% of the full dataset. This confirms that the problem is highly imbalanced and that accuracy alone would not be a reliable measure.

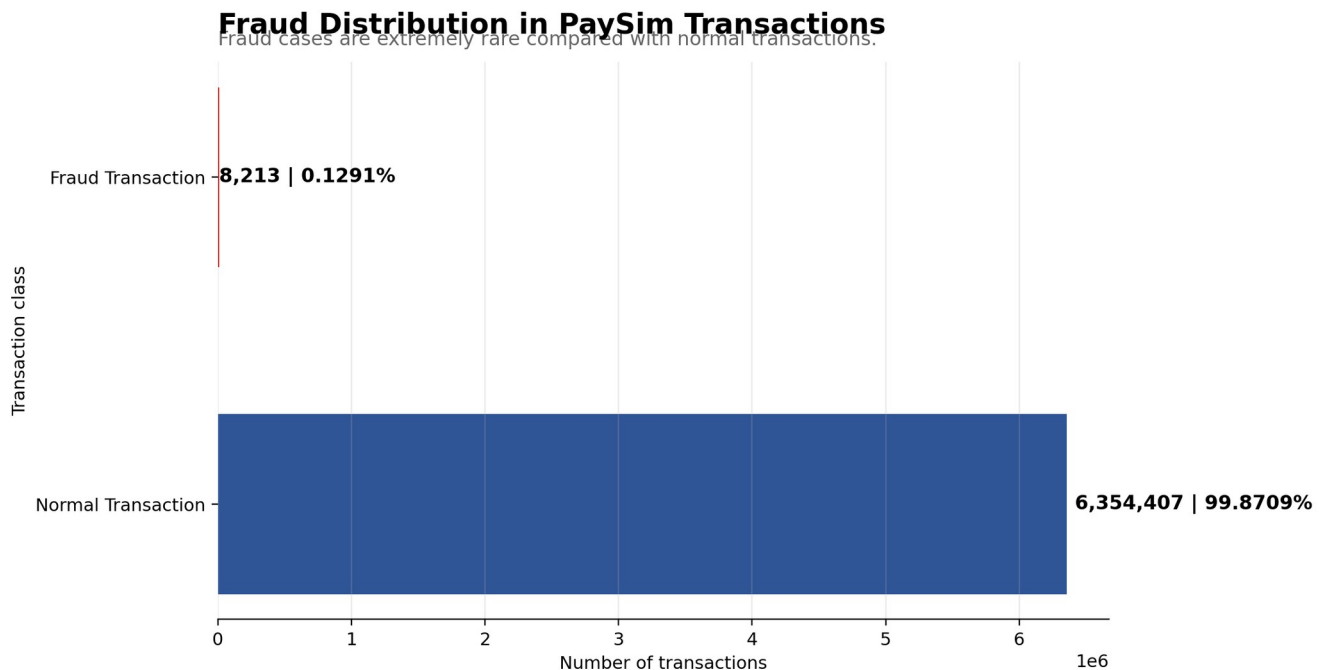


Figure 1. Full dataset fraud distribution.

3. Theory A-Z: Autoencoder-Based Anomaly Detection

3.1 What is anomaly detection?

Anomaly detection means finding records that look different from normal behaviour. In this notebook, a suspicious transaction is a transaction that does not match common normal transaction patterns. An anomaly does not automatically prove fraud; it means the record should be investigated.

3.2 What is an autoencoder?

An autoencoder is a neural network that learns to reconstruct its input. It has an encoder, a bottleneck, and a decoder. The encoder compresses the input, the bottleneck stores the compact representation, and the decoder rebuilds the original features.

3.3 Why can autoencoders detect anomalies?

If the model is trained mostly or only on normal data, it learns normal patterns well. Normal records are reconstructed with low error. Unusual records are harder to reconstruct and can produce high reconstruction error.

3.4 What is reconstruction error?

Reconstruction error is the difference between the original input features and the reconstructed output features. In this notebook, mean squared error is calculated for each transaction. A higher error means the model struggled to rebuild that transaction.

3.5 What is a threshold?

A threshold converts reconstruction error into a business decision. If reconstruction error is above the threshold, the transaction is flagged as suspicious. If it is below the threshold, the transaction is treated as normal.

3.6 Why train only on normal transactions?

The goal is to teach the autoencoder what normal behaviour looks like. If fraud is included heavily during training, the model may learn fraud patterns too, which weakens anomaly detection.

3.7 How are labels used?

The fraud label is not used as an input feature or training target. It is kept separately and used only after reconstruction error has been calculated, so model performance can be evaluated.

3.8 What are false positives and false negatives?

A false positive is a normal transaction wrongly flagged as suspicious. A false negative is a fraud transaction missed by the model. In fraud detection, false negatives can create financial risk, while false positives create investigation workload.

Concept	Simple Meaning
Normal transaction	Common transaction pattern learned by the model.
Suspicious transaction	Transaction with high reconstruction error.
Input = target	The autoencoder learns to reconstruct the same feature vector it receives.
Reconstruction error	How badly the model rebuilt the transaction.
Threshold	Cutoff used to decide normal vs suspicious.

4. Clean Top-to-Bottom Workflow Flowchart

The notebook follows a clear anomaly detection workflow from raw transactions to fraud evaluation.

Autoencoder-Based Anomaly Detection Workflow

The model learns normal transaction behaviour and flags high reconstruction-error transactions as suspicious.

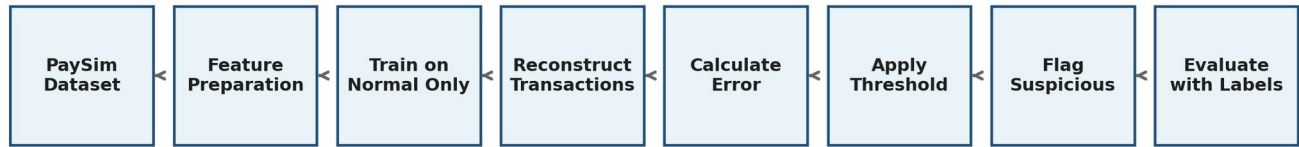


Figure 2. End-to-end anomaly detection workflow.

5. Variable and Method Flow

This section shows how the key variables move through the notebook. The naming is intentionally simple and avoids vague names such as temp or full.

Variable and Method Flow

How data moves from raw transactions to anomaly flags.

transactions	Raw PaySim data loaded from CSV
working_data	100,000 normal transactions + all 8,213 fraud transactions
features / labels	Model inputs separated from isFraud evaluation label
train_features	Normal transactions only for training
val_features	Normal transactions only for validation threshold
test_features	Normal + fraud transactions for final evaluation
train_processed	Scaled and one-hot encoded model input
autoencoder	Learns to reconstruct normal transaction features
reconstruction_error	Anomaly score for each test transaction
predicted_suspicious	Threshold-based suspicious transaction flag

Figure 3. Variable and method flow from raw data to anomaly flag.

Variable	Created From	Purpose
transactions	CSV file	Raw PaySim dataset.
working_data	normal_sample + fraud_sample	Practical working dataset for local training.
features	working_data without excluded columns	Model input columns.
labels	working_data[isFraud]	Evaluation labels only.
normal_data	working_data where isFraud = 0	Normal transaction pool.

fraud_data	working_data where isFraud = 1	Fraud records kept for test evaluation.
train_features	normal_train_data[feature_columns]	Normal-only training input.
val_features	normal_val_data[feature_columns]	Normal-only validation input.
test_features	test_data[feature_columns]	Final test input with normal + fraud.
train_processed	preprocessor.fit_transform(train_features)	Processed model input for training.
test_reconstruction_error	mean squared difference between test_processed and reconstruction	Anomaly score.
predicted_suspicious	reconstruction_error > threshold	Final suspicious transaction flag.

6. Data Preparation

6.1 Working Sample

The full dataset has more than 6.3 million rows. To keep local notebook training practical, a working sample was created while keeping all known fraud cases.

Transaction Type	Strategy	Count
Normal transactions	Random sample	100,000
Fraud transactions	Keep all fraud cases	8,213
Working dataset total	Combined and shuffled	108,213

6.2 Feature Selection

Used Features	Excluded Columns
step, type, amount, oldbalanceOrg, newbalanceOrig, oldbalanceDest, newbalanceDest	isFraud, isFlaggedFraud, nameOrig, nameDest

6.3 Train/Validation/Test Split

Dataset	Content	Shape	Purpose
Training	Normal only	70,000 rows, 11 processed features	Model learns normal behaviour.
Validation	Normal only	15,000 rows, 11 processed features	Training check and threshold reference.
Test	Normal + fraud	23,213 rows, 11 processed features	Final fraud detection evaluation.

6.4 Preprocessing

A ColumnTransformer was used so numeric and categorical columns receive the right transformation. The transformer was fitted only on training data and then applied to validation and test data to avoid leakage.

Input Type	Columns	Transformation	Output
Numeric	step, amount, balances	StandardScaler	6 scaled numeric columns
Categorical	type	OneHotEncoder	5 transaction-type columns
Final input	All processed features	Combined matrix	11 model input features

7. Model Architecture and Training

7.1 Architecture

Layer	Output Size	Purpose
Input	11	Receives processed transaction features.
Encoder Dense	8	Learns compressed transaction patterns.
Bottleneck	4	Stores compact hidden representation.
Decoder Dense	8	Rebuilds hidden representation.
Output	11	Reconstructs original processed features.

The model was compiled with Adam optimizer and mean squared error loss. For the autoencoder, the input and target are the same processed transaction features.

7.2 Training Results

Metric	Final Value	Interpretation
Training loss	0.013953	Reconstruction loss on normal training transactions.
Validation loss	0.013549	Reconstruction loss on normal validation transactions.

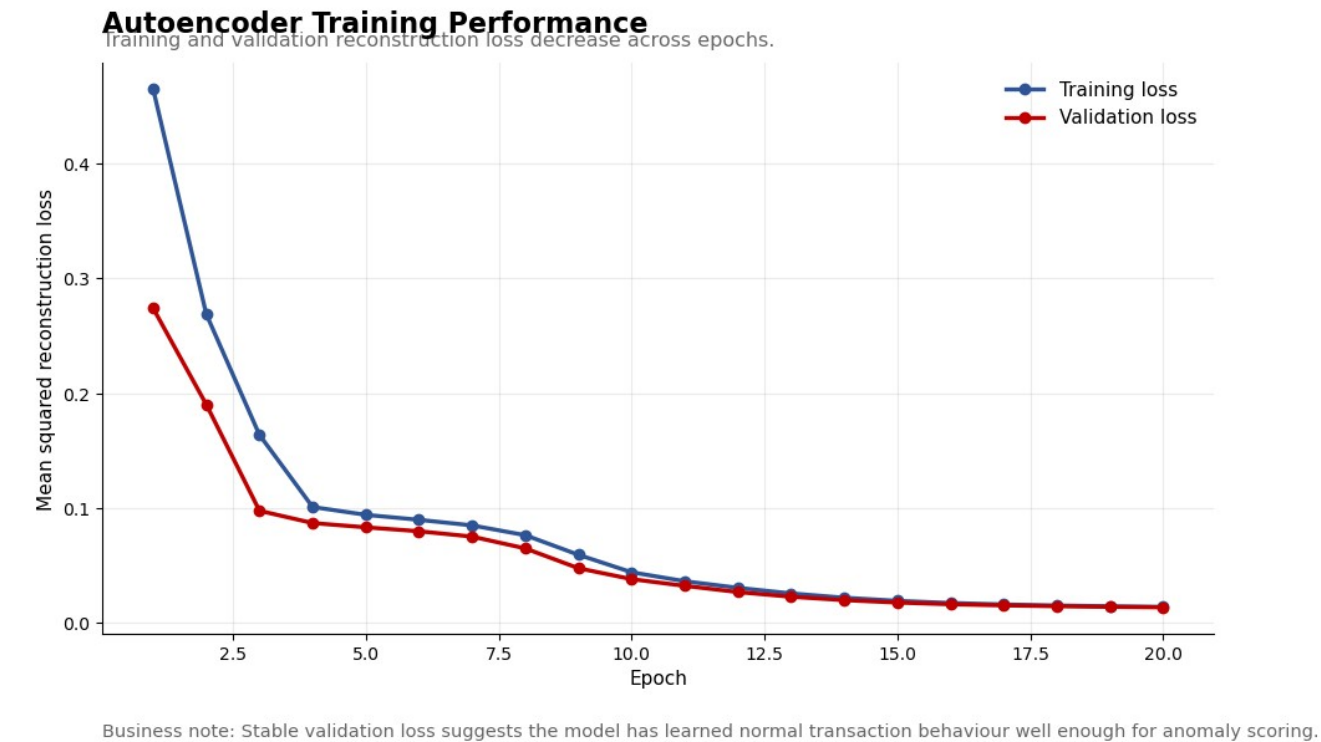


Figure 4. Autoencoder training and validation loss.

The training and validation losses decreased and stayed close, suggesting that the model learned normal transaction behaviour without obvious overfitting.

8. Reconstruction Error Results and Chart Interpretation

After training, the autoencoder reconstructed the test set and reconstruction error was calculated for each transaction.

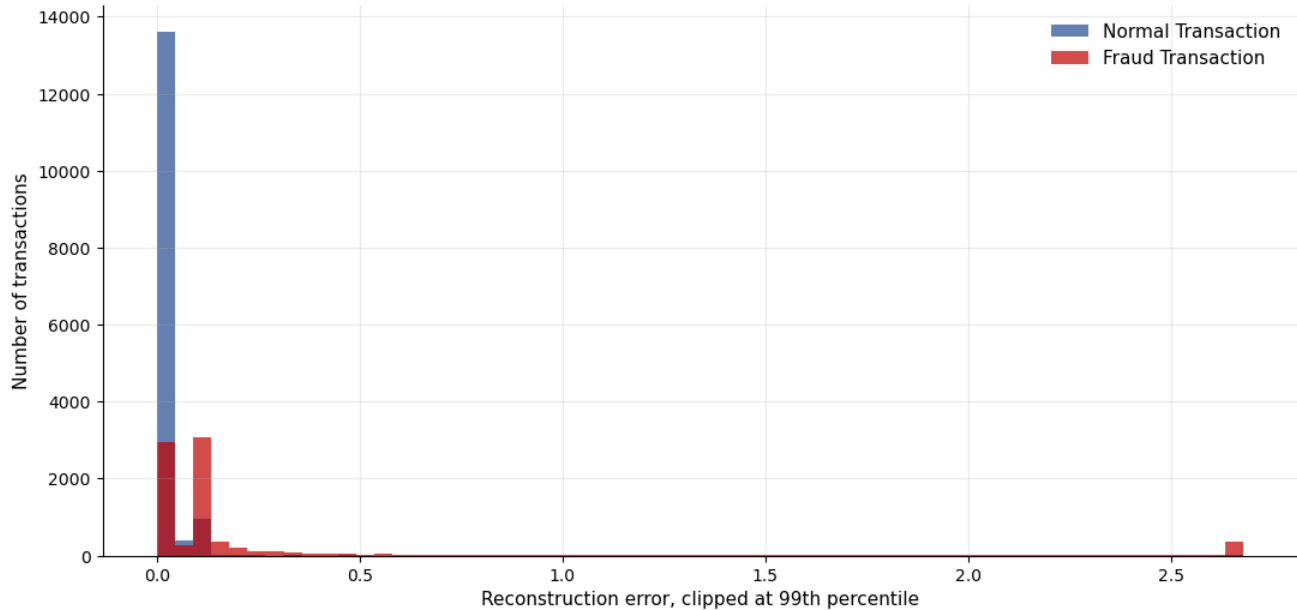
Transaction Type	Count	Mean Error	Median Error	Max Error
Normal Transaction	15,000	0.013375	0.002064	3.364668
Fraud Transaction	8,213	0.550486	0.110238	28.858944

Key interpretation

Fraud transactions had much higher mean and median reconstruction error than normal transactions. This supports the anomaly detection logic that unusual transactions are harder to reconstruct.

Reconstruction Error Distribution by Transaction Class

Fraud transactions generally show higher reconstruction error than normal transactions.



Business note: Higher reconstruction error can be used as an anomaly score for suspicious transaction detection.

Figure 5. Reconstruction error distribution by transaction class.

Most normal transactions are concentrated near very low reconstruction error. Fraud transactions are more spread out and have a longer high-error tail. There is still overlap, so reconstruction error is useful but not perfect.

9. Thresholding and Evaluation

9.1 Selected Threshold

Metric	Value
Mean validation reconstruction error	0.013549
Median validation reconstruction error	0.002067
95th percentile threshold	0.098729

The selected threshold was the 95th percentile of normal validation reconstruction error. This means transactions with reconstruction error greater than 0.098729 were flagged as suspicious.

9.2 Threshold Trade-Off

Percentile	Threshold	Flagged Transactions	Flagged Share	Precision	Recall	F1
90.0	0.031466	6,935	29.876%	0.781	0.660	0.715
92.5	0.084414	6,106	26.304%	0.823	0.612	0.702

95.0	0.098729	5,721	24.646%	0.868	0.605	0.713
97.5	0.107839	5,116	22.039%	0.925	0.576	0.710
99.0	0.118975	2,336	10.063%	0.933	0.265	0.413

Fraud Detection Threshold Trade-Off

Higher thresholds improve precision but reduce recall.

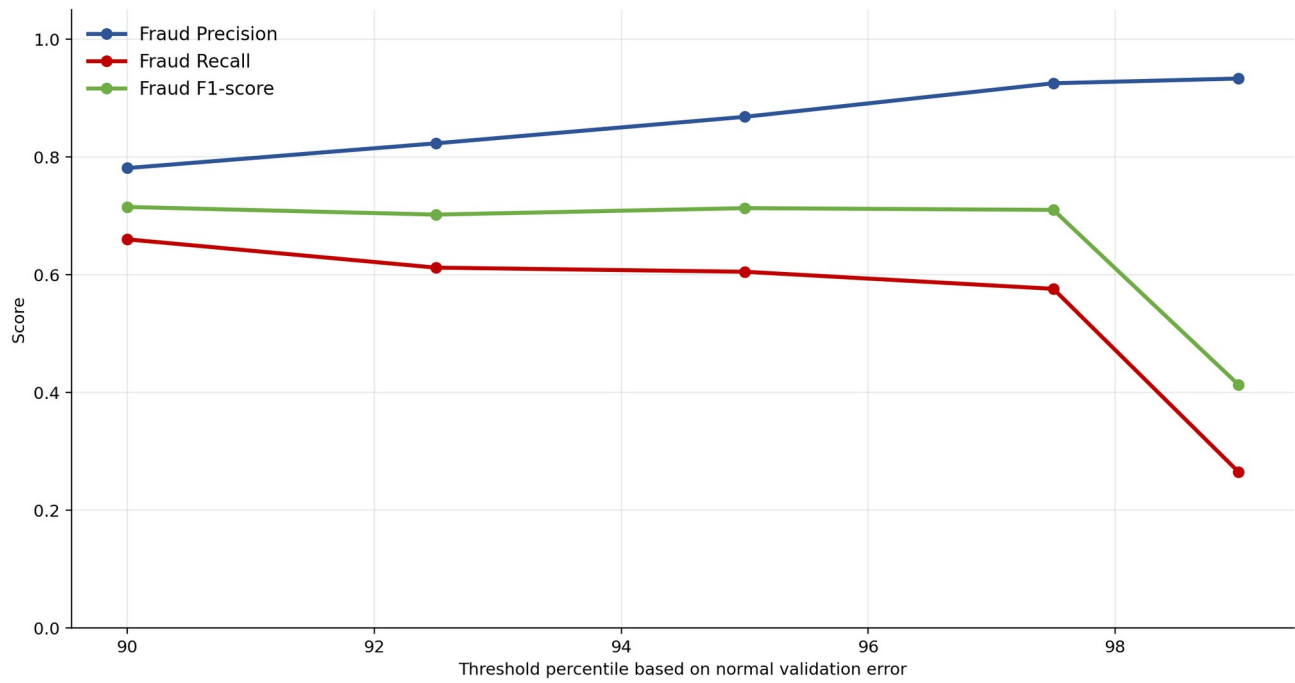


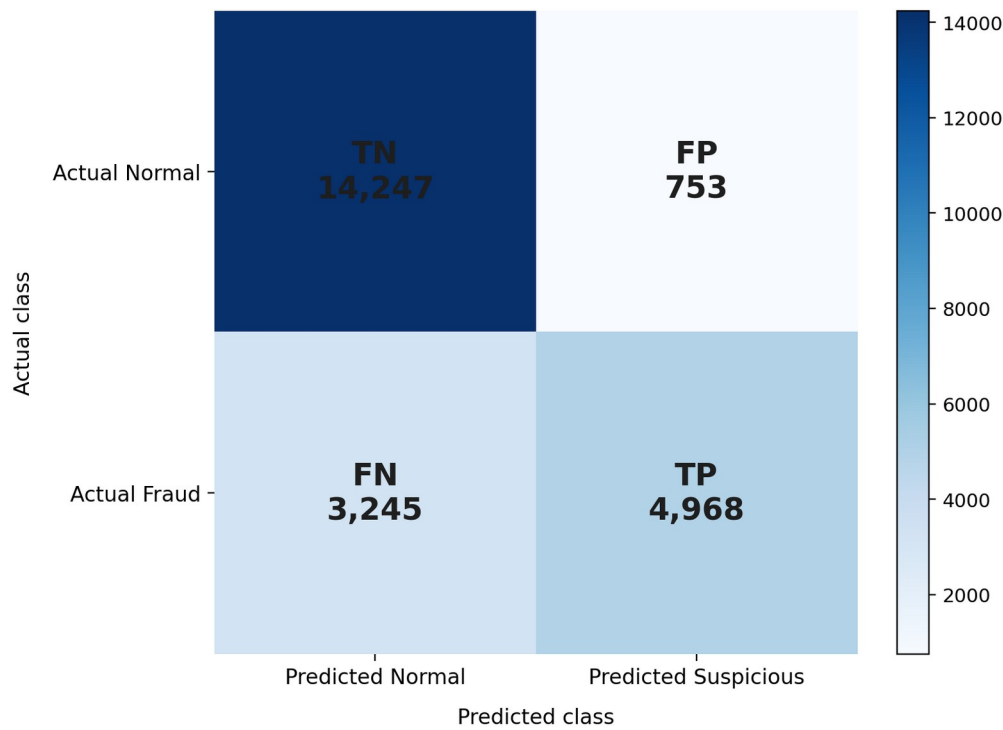
Figure 6. Threshold precision-recall trade-off.

As the threshold increases, fraud precision improves but recall decreases. The 99th percentile threshold is too strict because recall falls sharply. The 95th percentile threshold is a balanced choice for this notebook.

9.3 Confusion Matrix

Confusion Matrix: Autoencoder Fraud Detection

Selected threshold: 95th percentile of normal validation reconstruction error.



Business note: TP = detected fraud, FN = missed fraud, FP = false alert, TN = correctly ignored normal transaction.

Figure 7. Confusion matrix for 95th percentile threshold.

Term	Count	Meaning
TN	14,247	Normal transactions correctly predicted as normal.
FP	753	Normal transactions wrongly flagged as suspicious.
FN	3,245	Fraud transactions missed by the model.
TP	4,968	Fraud transactions correctly flagged as suspicious.

10. Business KPI Summary

KPI	Value	Business Meaning
Detected Fraud Transactions	4,968	Fraud cases correctly flagged as suspicious.
Missed Fraud Transactions	3,245	Fraud cases not detected by current threshold.
False Alerts	753	Normal transactions wrongly sent for investigation.
Correctly Ignored Normal Transactions	14,247	Normal transactions correctly left alone.

Fraud Precision	0.868	Of suspicious alerts, 86.8% were actual fraud.
Fraud Recall	0.605	The model detected 60.5% of actual fraud.
Fraud F1-score	0.713	Balance between precision and recall.
False Positive Rate	0.050	Around 5.0% of normal transactions were falsely flagged.
Missed Fraud Rate	0.395	Around 39.5% of fraud was missed.

Business reading

The selected threshold creates high-quality suspicious alerts, but it still misses a meaningful share of fraud. If the business prioritizes catching more fraud, the threshold should be lowered. If the business prioritizes fewer false alerts, the threshold should be raised.

11. Limitations and Next Steps

11.1 Limitations

- The working dataset was sampled for local training, so evaluation distribution differs from the full business distribution.
- The selected threshold is simple and explainable, but not necessarily optimal.
- Reconstruction error is useful but not perfect because normal and fraud error distributions overlap.
- The model uses a simple dense autoencoder architecture; stronger architectures may improve recall.
- The notebook excludes customer/account IDs directly, so network-style fraud behaviour is not captured.

11.2 Recommended Next Steps

- Test threshold selection based on business cost: cost of false positive vs cost of missed fraud.
- Try deeper autoencoder architectures and compare reconstruction error separation.
- Use early stopping and callbacks for more robust training.
- Profile high-error fraud and high-error normal transactions to understand business patterns.
- Compare autoencoder anomaly detection with supervised fraud classifiers.
- Create a final executive BI dashboard showing alerts, detected fraud, missed fraud, and threshold trade-offs.

12. Interview-Style Explanation

In this project, I used an autoencoder for anomaly detection on financial transaction data. The goal was to learn normal transaction behaviour and then detect suspicious transactions using reconstruction error.

I first prepared the PaySim dataset and created a practical working sample with 100,000 normal transactions and all available fraud transactions. I excluded fraud labels and ID columns from model input to avoid leakage. Numeric features were scaled and transaction type was one-hot encoded using a ColumnTransformer.

The autoencoder was trained only on normal transactions. During training, the model learned to reconstruct normal transaction behaviour. After training, I reconstructed the test set and calculated mean squared reconstruction error for each transaction.

Fraud transactions had much higher reconstruction error than normal transactions, which supported the anomaly detection approach. I selected the 95th percentile of validation reconstruction error as the anomaly threshold. At this threshold, the model achieved fraud precision of 0.868 and fraud recall of 0.605.

From a business point of view, the model creates useful suspicious transaction alerts, but it also misses some fraud. Threshold tuning is therefore a business decision based on whether the company wants to catch more fraud or reduce false alerts.

13. Glossary

Term	Simple Meaning
Anomaly	A record that looks unusual compared with normal behaviour.
Autoencoder	A neural network that learns to reconstruct its input.
Encoder	Part of the model that compresses input features.
Bottleneck	Small hidden layer that stores compact representation.
Decoder	Part of the model that rebuilds the input from the bottleneck.
Reconstruction error	Difference between original input and reconstructed output.
Threshold	Cutoff used to decide normal vs suspicious.
Precision	Of predicted suspicious cases, how many were truly fraud.
Recall	Of actual fraud cases, how many were detected.
F1-score	Balance between precision and recall.
True Positive (TP)	Fraud correctly flagged as suspicious.
False Positive (FP)	Normal transaction wrongly flagged as suspicious.
False Negative (FN)	Fraud transaction missed by the model.
True Negative (TN)	Normal transaction correctly predicted as normal.
Data leakage	When information from labels/test data accidentally influences training.
ColumnTransformer	Scikit-learn tool that applies different preprocessing to different column groups.
StandardScaler	Scales numeric features to comparable ranges.
OneHotEncoder	Converts categorical values into numeric indicator columns.

14. Final Notebook Status

The core anomaly detection workflow is complete. The notebook successfully demonstrates how an autoencoder can support suspicious transaction detection by learning normal behaviour and using reconstruction error as an anomaly score.

Completion note

Optional final notebook polish can still be added later: final business interpretation, limitations, next steps, and interview explanation inside the notebook. This master document already captures those elements for documentation.